

Reverse-check review package — B-L-SM_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	B-L-SM/repair2/final.fr
original model name	B-L-SM_gen (hidden from the agent)
paper	B-L-SM/text/1811.11452.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the .fr)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LZpKin (:=)

```
Block[{mu,nu}, -1/4 FS[Zp,mu,nu] FS[Zp,mu,nu]]
```

LChi (:=)

```
Block[{mu,ii}, ExpandIndices[(del[HC[Chi],mu] - I*2*g1p*Zp[mu]*HC[Chi])*(del[Chi,mu] +  
↪ I*2*g1p*Zp[mu]*Chi) - muChi2*HC[Chi]*Chi - lambda2BL*(HC[Chi]*Chi)^2 -  
↪ lambda3BL*(Phibar[ii]*Phi[ii])*(HC[Chi]*Chi), FlavorExpand -> SU2D]]
```

LBLCurrent (:=)

```
Block[{mu}, ExpandIndices[-g1p*Zp[mu]*(1/3 QLbar.Ga[mu].QL + 1/3 uRbar.Ga[mu].uR + 1/3  
↪ dRbar.Ga[mu].dR - LLbar.Ga[mu].LL - lRbar.Ga[mu].lR - nuRbar.Ga[mu].nuR), FlavorExpand  
↪ -> {SU2D, Generation, Colour}]]
```

YMRRules (=)

```
{YMRMat[1,1] -> YMR1x1, YMRMat[1,2] -> YMR1x2, YMRMat[1,3] -> YMR1x3, YMRMat[2,1] -> YMR2x1,  
↪ YMRMat[2,2] -> YMR2x2, YMRMat[2,3] -> YMR2x3, YMRMat[3,1] -> YMR3x1, YMRMat[3,2] ->  
↪ YMR3x2, YMRMat[3,3] -> YMR3x3}
```

YNURules (=)

```
{YNUMat[1,1] -> YNU1x1, YNUMat[1,2] -> YNU1x2, YNUMat[1,3] -> YNU1x3, YNUMat[2,1] -> YNU2x1,  
↪ YNUMat[2,2] -> YNU2x2, YNUMat[2,3] -> YNU2x3, YNUMat[3,1] -> YNU3x1, YNUMat[3,2] ->  
↪ YNU3x2, YNUMat[3,3] -> YNU3x3}
```

LNuYukNonHC (:=)

```
Block[{sp,ff1,ff2}, ExpandIndices[-YNUMat[ff1,ff2] (LLbar[sp,1,ff1].nuR[sp,ff2] (Phibar[2] -  
↪ vev/Sqrt[2]) - LLbar[sp,2,ff1].nuR[sp,ff2] Phibar[1]) - 1/2 YMRMat[ff1,ff2]  
↪ anti[CC[nuR]] [sp,ff1].nuR[sp,ff2] Chi, FlavorExpand -> Generation] /. YNURules /.  
↪ YMRRules]
```

LNuYuk (:=)

LNuYukNonHC + HC[LNuYukNonHC]

LScalarDiag (:=)

```
Block[{mu}, ca*sa*del[H,mu]*del[H2,mu] - ca*sa*muChi2*H*H2 - ca*sa*lambda3BL*vev^2*H*H2/2 +  
↪ ca*lambda3BL*vev*xBL*H*H2 - 3*ca*sa*lambda2BL*xBL^2*H*H2]
```

LTot (:=)

LZpKin + LChi + LBLCurrent + LNuYuk + LScalarDiag

Blank-slate reconstruction

(reconstruction phase failed — see agent_runs in the tool result)

Paper cross-check

(not run — provide paper_tex_path to enable the term-by-term comparison)

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 32 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: [] approve [] approve with corrections [] reject
- Notes: